

Experiment 4: Crypt Arithmetic Problem (Python)

Aim

To write a Python program to solve the **Crypt Arithmetic Problem**.

Algorithm

1. Assign digits to the letters.
2. Check whether the arithmetic equation is correct.
3. If the equation is satisfied, print the solution.
4. Otherwise, print "No Solution".

Program

S, E, N, D = 9, 5, 6, 7

M, O, R, Y = 1, 0, 8, 2

$SEND = S*1000 + E*100 + N*10 + D$

$MORE = M*1000 + O*100 + R*10 + E$

$MONEY = M*10000 + O*1000 + N*100 + E*10 + Y$

```
print("SEND =", SEND)
```

```
print("MORE =", MORE)
```

```
print("MONEY =", MONEY)
```

```
if SEND + MORE == MONEY:
```

```
    print("Solution Found")
```

```
else:
```

```
    print("No Solution")
```

Input

Not required

Output

```
SEND = 9567  
MORE = 1085  
MONEY = 10652  
Solution Found
```

Result

Thus, the Python program to solve the Crypt Arithmetic Problem was executed successfully, and the correct solution was obtained.